NO KEYBOARD · NO MOUSE · NO CONTROLLER

ECHO

You are blind in a cave.
Your voice is your only light.

A horror escape game played in total darkness, controlled only by your voice and your head. A webcam and a microphone are the whole controller.



In-game artwork: a Hollow. Blind. It hunts by sound.

TEAM	TRACK	HACKATHON
Solo Levelling	Track 2: PlayLab	MGM2026

PROPOSED SOLUTION

Sound is sight. The input is not a button in disguise: it IS the core mechanic.

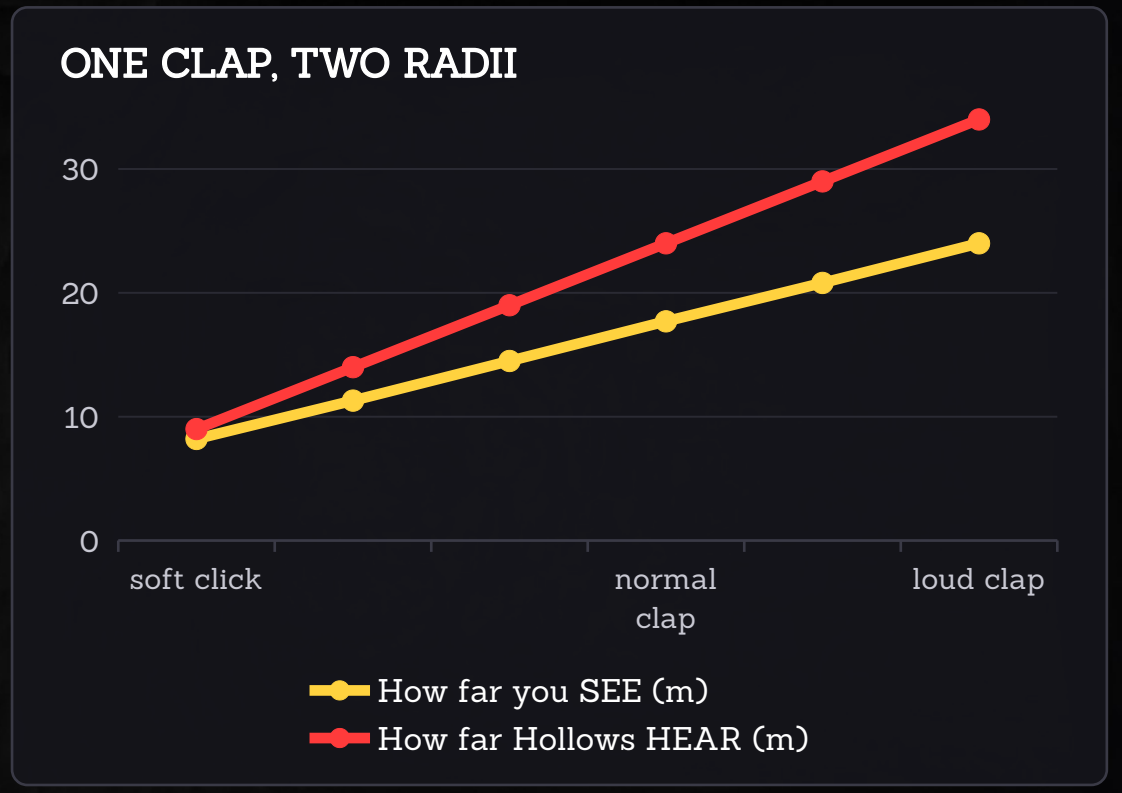
THE CHALLENGE Play without keyboard, mouse, touch or controller, where the input is meaningfully tied to gameplay.

CLAP · CLICK · KNOCK
A sonar pulse reveals the cave. Louder = see further, and be heard further.

HUM
A quiet beam where you look. Pitch sets width, loudness sets range. Heard only within ~5 m.

TURN YOUR HEAD
Look left and right. MediaPipe tracks head yaw.

LEAN IN · LEAN BACK
Walk. A gentle lean is silent; a full lean makes footsteps they can hear.

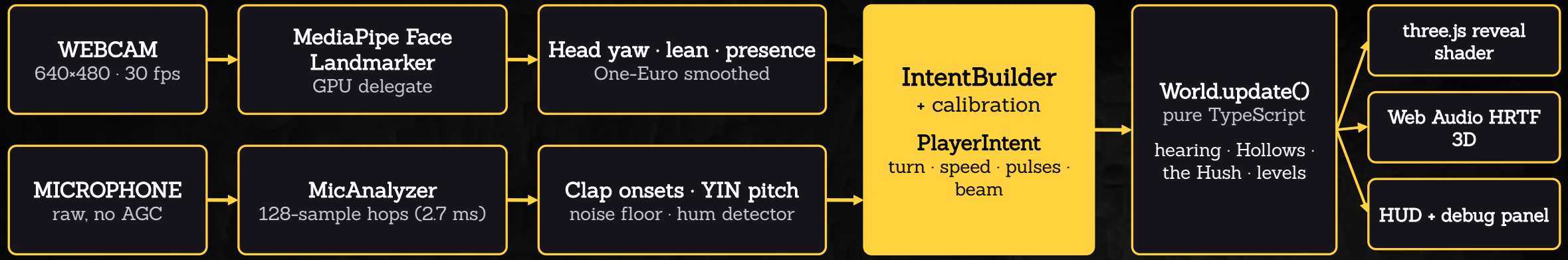


USP: every input is a bet. Sound always carries further than it reveals, so seeing the cave tells the Hollows where you are.

TECHNICAL APPROACH

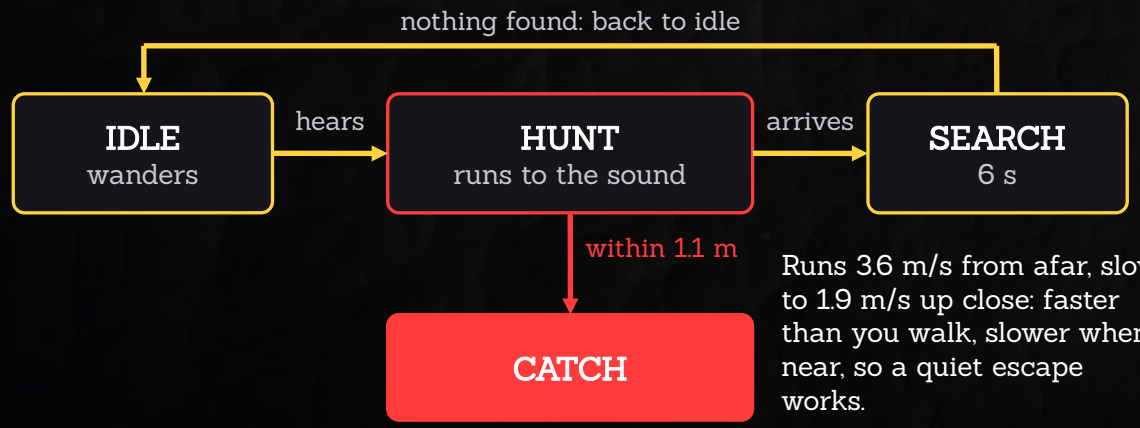
ECHO

Sensor → signal processing → intent → pure game logic → real-time feedback, all in the browser.



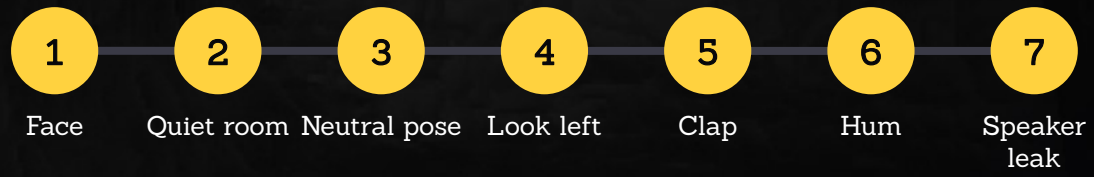
The game logic never touches a sensor. A dev-only keyboard sim feeds the same PlayerIntent, which is how 141 unit tests drive the game.

THE HOLLOW'S BRAIN



Runs 3.6 m/s from afar, slows to 1.9 m/s up close: faster than you walk, slower when near, so a quiet escape works.

HANDS-FREE CALIBRATION · ~45 s



Every step: says what and why → countdown → live reading → result.

TypeScript · Vite · three.js · MediaPipe · Web Audio · Vitest

FEASIBILITY & VIABILITY

It already works: a playable 3-level prototype, measured on an ordinary laptop with integrated graphics.

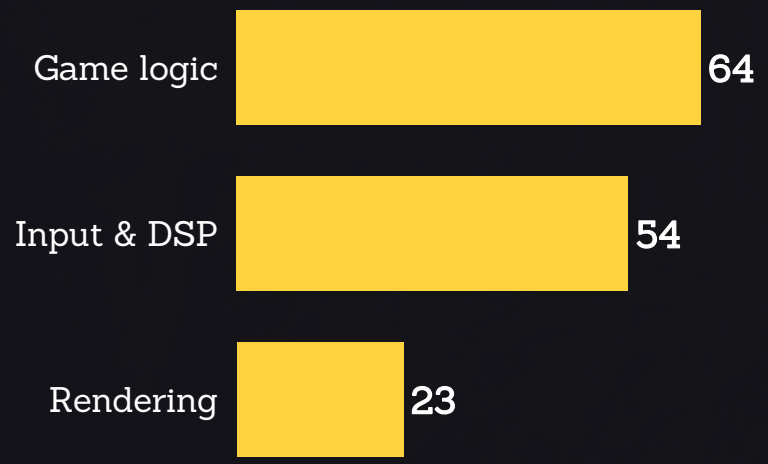
30 fps
face tracking on an Intel integrated GPU

6.5 ms
to render the cave per frame, inside a 20 ms budget

141
unit tests, all passing

0
extra hardware: webcam + mic, runs offline in Chrome

141 TESTS BY LAYER



RISKS WE DESIGNED FOR

Noisy hackathon hall	Calibrated noise floor; a clap must spike, then drop 8 dB within 75 ms
The game hears itself	Headphones + a speaker-leak test during calibration
Face lost, bad light	Auto-pause after 0.4 s without a face
Weak laptop	Adaptive render scale below 40 fps; all assets bundled offline

Viable: browser distribution, zero install, zero hardware cost. Levels are ASCII maps, so new content is cheap.

IMPACT & BENEFITS

Your body is the controller.

Accessibility

Fully hands-free play for people who can't use a keyboard, mouse or controller.

Input becomes the game

Alt-input as the fun, not a gimmick: your noise is both your light and your danger.

Real immersion

You hold your breath, lean in slowly and hum quietly. The fear is physical.

Reusable & low barrier

A face + voice → intent pipeline any browser game can adopt. Any laptop, no install.



Seen only in the flash of your own clap.

RESEARCH & REFERENCES

The techniques under the hood, plus related research and games.

Signal processing

- YIN pitch estimation: de Cheveigné & Kawahara, JASA 2002
- One Euro Filter: Casiez, Roussel & Vogel, CHI 2012
- Percussive onset detection with an adaptive noise floor

Vision & rendering

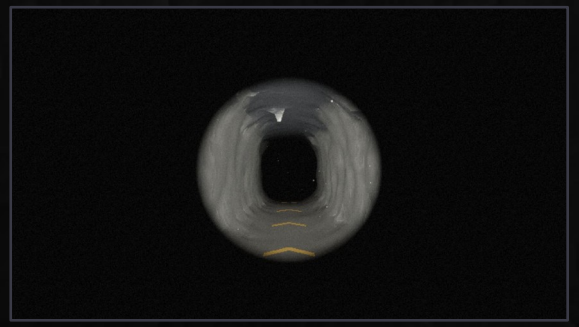
- MediaPipe Face Landmarker (Google)
- Surface Nets meshing of distance fields: Gibson, 1998
- three.js · Web Audio API (HRTF panning)

Related work

- Human echolocation: Thaler, Arnott & Goodale, PLoS ONE 2011
- Sound-driven horror games: Dark Echo (2015), Perception (2017), Stifled (2017)



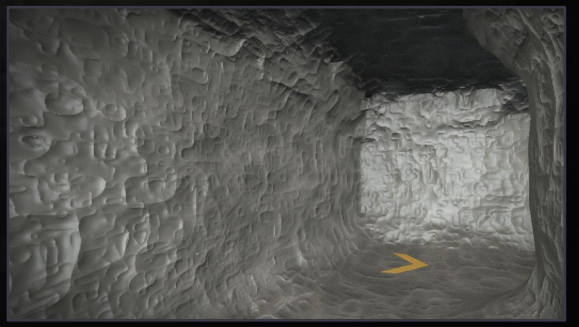
Clap: the pulse reveals a Hollow



Hum: a narrow, quiet beam



It heard you



Awakening: gold arrows to the exit